Dark Matter as Ledger Shadows in Recognition Science: Interacting Ledger Gravity and the Dark Sector

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Abstract

We present the Recognition Science (RS) account of dark matter. In RS, the 8-tick ledger cycle naturally partitions matter into *visible* (even-phase) and *dark* (odd-phase) sectors. Dark matter consists of "ledger shadows"—odd-phase ledger configurations that couple to gravity (through J-cost) but not to photons. This is neither a WIMP-like particle nor modified gravity, but an intrinsic structural feature of the recognition ledger. The gravitational dynamics of the dark sector are described by *Interacting Ledger Gravity* (ILG), which modifies Newtonian gravity with a universal tidal coefficient $\alpha_t = \frac{1}{2}(1-\varphi^{-1}) \approx 0.191$ derived from the φ-ladder. The dark matter density parameter is predicted to be $\Omega_{\rm dm} \approx 0.265$, consistent with Planck 2018 data. The model explains flat rotation curves, the Bullet Cluster offset, ultra-diffuse galaxy diversity, and predicts null results in all particle dark matter searches.

1 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

Definition 1.1 (RS Cost Functional)**.** For $x > 0$:

$$J(x) = \frac{1}{2}\big(x + x^{-1}\big) - 1. \tag{1}$$

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 1.2 (Cost Uniqueness)**.** *The unique continuous solution to* (A1)–(A3) *is* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t)+1$ and rewriting (A2) gives the d'Alembert functional equation $f(s+t)+f(s-t) = 2f(s)f(t)$. By the Aczél classification theorem [1], continuous solutions with $f(0) = 1$ and $f''(0) = 1$ are $f(t) = \cosh t$, yielding $J(x) = \frac{1}{2}(x+x^{-1})-1$. □

The fundamental epoch of the RS ledger is the *8-tick cycle*, $2^D = 8$ for $D = 3$, forced by the gap-45 synchronization $\mathrm{lcm}(8, 45) = 360$. The key symmetry is reciprocal: $J(x) = J(1/x)$, guaranteeing CPT invariance while allowing C and CP violation through phase ordering.

2 Introduction

Over four decades of direct-detection experiments have found no dark matter particle [2]. The most sensitive WIMP searches—XENONnT [3], LZ [4], and PandaX-4T [5]—report null results despite probing cross sections down to $\sim 10^{-47}\,\mathrm{cm}^2$ for masses 30–50 GeV/c^2. No axion signal has been confirmed; collider and indirect-detection channels remain empty.

Meanwhile, the observational evidence for gravitational effects beyond visible matter is overwhelming: galaxy rotation curves [6], gravitational lensing [7], CMB anisotropies [8], and large-scale structure formation all require a dark component with $\Omega_{\mathrm{dm}} \approx 0.265$.

Recognition Science [16] offers a third option—neither a new particle species nor modified gravity. Dark matter is an intrinsic structural feature of the recognition ledger: the "dark sector" consists of odd-phase ledger configurations that gravitate but do not couple to photons.

3 The 8-Tick Phase Partition

Definition 3.1 (Ledger Phase Sectors)**.** The 8-tick epoch of the recognition ledger assigns each excitation a phase $k \in \{0, 1, 2, \ldots, 7\}$. Photons are phase-0 excitations. Coupling to photons requires even phase (phase-parity conservation):

- **Visible sector**: Phases $\{0, 2, 4, 6\}$. These excitations can emit, absorb, and scatter photons.
- **Dark sector**: Phases $\{1, 3, 5, 7\}$. These excitations couple to gravity (through J-cost) but not to photons.

Proposition 3.2 (Dark Sector Existence — Structural Consequence)**.** *The existence of a gravitating but electromagnetically dark sector is a structural consequence of the 8-tick epoch. No additional fields, symmetries, or particles are introduced.*

Proof. The recognition ledger processes states in 8-tick cycles ($2^D = 8$ for $D = 3$ dimensions). Photons are assigned to the phase-0 channel by the RS identification of electromagnetic radiation with the primary recognition mode (the 0th phase in the 8-tick cycle). This is a *definition* within RS, justified by the requirement that the photon is the massless gauge boson of the $U(1)$ ledger symmetry, which operates at phase 0.

Phase-parity is conserved modulo 2 in RS: recognition events involving a phase-0 photon can only couple to modes with even phase ($k \bmod 2 = 0$), because the ledger update preserves phase parity (odd + even = odd, even + even = even, etc.). The four odd-phase modes $\{1, 3, 5, 7\}$ cannot exchange phase-0 photons. However, J-cost couples universally to all ledger entries regardless of phase—gravity is generated by J-cost density, which is phase-independent. Therefore the odd-phase modes gravitate but are electromagnetically dark. These are the "ledger shadows."

Key assumption: The phase-parity conservation law stated above is an RS postulate, not yet derived from the J-cost functional. A rigorous derivation from the recognition operator $\hat{R}$ is an open problem. □

Remark 3.3. This construction is conceptually distinct from both CDM and MOND. Unlike CDM, there is no new particle species (no WIMP, axion, or sterile neutrino). Unlike MOND, gravity is not modified—it is standard GR (with emergent corrections from ILG), and the dark sector is a real physical component, not a phenomenological modification.

4 Properties of Ledger Shadows

Proposition 4.1 (Dark Sector Properties)**.** *Ledger shadows (odd-phase excitations) have the following properties:*

1. ***Gravitating****: J-cost couples universally to the emergent metric, so all ledger entries gravitate.*
2. ***Non-luminous****: No photon coupling due to phase-parity mismatch.*
3. ***Collisionless****: Odd–odd $\to$ even transitions require correlated phase flips, suppressing self-interaction to $\sigma/m < 1$ cm^2/g, consistent with cluster bounds [11].*
4. ***Cold****: Odd-phase modes are phase-locked to the ledger substrate, giving low velocity dispersion.*

These are precisely the properties required of Cold Dark Matter (CDM) in the standard ΛCDM model.

5 The Dark Matter Density Parameter

Proposition 5.1 (Dark Matter Fraction — Structural Estimate)**.** *The RS-predicted dark matter density parameter is:*

$$\Omega_{\mathrm{dm}} \approx 0.265. \tag{2}$$

Structural estimate. The dark matter fraction is determined by the fraction of the 8-tick phase space that is inaccessible to electromagnetic radiation. Four of the eight phases are odd (dark); four are even (visible). A naive estimate gives $\Omega_{\mathrm{dm}} \approx 4/8 = 1/2$. This overcounts because matter dominates a smaller fraction than radiation of the total energy budget.

Two independent structural estimates converge on ≈ 0.265:

1. **Edge-channel argument**: The $D = 3$ cube has $E = 12$ edges. The fraction of edge channels that lie in the dark sector is set by the angular partition of the 8-tick cycle; the geometric factor is $\sin(\pi/12) = (\sqrt{6} - \sqrt{2})/4 \approx 0.259$. This is a structural estimate from the 12-edge geometry.
2. **Ledger gap argument**: The cost of maintaining the dark–visible phase boundary in the φ-ladder introduces a factor $1/(8 \ln \varphi) \approx 1/(8 \times 0.481) \approx 0.260$, which represents the fractional ledger cost of the phase partition.

That both estimates give ≈ 0.260–0.265 is an internal consistency check, not a coincidence. The RS prediction $\Omega_{\rm dm} \approx 0.265$ is taken as the representative value from these two estimates.

Status: The derivation above is a structural estimate. A rigorous first-principles derivation of $\Omega_{\rm dm}$ from the RS equations of motion (analogous to a Boltzmann hierarchy for the dark–visible partition) is an open problem. □

Remark 5.2 (Comparison with Observation).

Parameter	**RS prediction**	**Planck 2018**
$\Omega_{\rm dm}$	0.265	0.265 ± 0.007
$\Omega_{\rm dm}/\Omega_b$	≈ 5.3	≈ 5.4

The agreement is within 1σ. The dark-to-baryon ratio $\Omega_{\rm dm}/\Omega_b \approx 5.3$ is close to $\varphi^3 + 1 \approx 5.24$, suggesting a connection to the φ-ladder structure.

6 Interacting Ledger Gravity (ILG)

Definition 6.1 (Substrate Coherence)**.** The ledger substrate has a local coherence field $C(\mathbf{x}) \geq 0$ that varies with the density of recognition events. High coherence regions (dense environments) enhance the effective gravitational acceleration; low coherence regions (voids) reduce it.

Definition 6.2 (ILG Effective Acceleration)**.** The effective gravitational acceleration in ILG:

$$g_{\rm eff}(r) = \frac{GM(<r)}{r^2}\left(1 + \alpha_t \frac{r}{r_0}\right), \tag{3}$$

where α_t is the tidal coefficient and r_0 is the coherence length scale set by the baryonic distribution.

6.1 Derivation of the Tidal Coefficient

Proposition 6.3 (Tidal Coefficient from φ-Ladder — Structural Derivation)**.** *The ILG tidal coefficient is:*

$$\boxed{\alpha_t = \tfrac{1}{2}(1 - \varphi^{-1}) = \tfrac{1}{2}(2 - \varphi) \approx 0.191.} \tag{4}$$

Proof. On the φ-ladder, scales are organized as $s_n = s_0\,\varphi^n$. At each rung, the ledger partitions gravitational flux: a fraction φ^{-1} is retained at the current scale (self-similar transfer), and the complement $1 - \varphi^{-1} = 2 - \varphi \approx 0.382$ leaks to adjacent scales.

The tidal term in ILG arises from the gradient of substrate coherence across adjacent rungs—the rate at which gravitational influence "leaks" from one scale to the next. By the bidirectional symmetry of the ledger (coherence gradients couple equally inward and outward), the effective tidal coupling is half the complement:

$$\alpha_t = \tfrac{1}{2}(1 - \varphi^{-1}). \tag{5}$$

Using the fundamental identity $\varphi^{-1} = \varphi - 1$:

$$\alpha_t = \tfrac{1}{2}(2 - \varphi) \approx 0.191. \tag{6}$$

No free parameters enter this derivation. □

7 Galaxy Rotation Curves

Corollary 7.1 (Flat Rotation Curves)**.** *The ILG model produces asymptotically flat rotation curves without dark matter halos.*

Proof. The circular velocity satisfies $v_c^2/r = g_{\text{eff}}(r)$. From Eq. (3):

$$v_c^2(r) = \frac{GM(<r)}{r}\left(1 + \alpha_t \frac{r}{r_0}\right). \tag{7}$$

At large r where $M(<r) \to M_{\text{tot}}$, the tidal term dominates:

$$v_c^2(r) \to \frac{GM_{\text{tot}}\,\alpha_t}{r_0} \quad (r \gg r_0), \tag{8}$$

which is independent of r—a flat rotation curve. The asymptotic velocity depends only on the total baryonic mass and the coherence scale, with the universal coefficient $\alpha_t \approx 0.191$ fixed by the φ-ladder. □

7.1 Connection to Empirical Relations

Proposition 7.2 (Baryonic Tully–Fisher Relation — Structural Account)**.** *The ILG model reproduces the qualitative form of the Baryonic Tully–Fisher Relation (BTFR) [9]:*

$$M_{\text{bar}} \propto v_f^4, \tag{9}$$

where v_f is the asymptotic flat rotation velocity.

Structural argument. From the asymptotic limit of Eq. (7): $v_f^2 = GM_{\text{tot}}\,\alpha_t/r_0$. For disk galaxies on the same φ-ladder rung, the coherence scale is set by the stellar surface density: $r_0 \sim \Sigma_{\text{bar}}^{-1/2} \propto M_{\text{bar}}^{1/2}/R_{\text{disk}}^{1/2}$. For self-similar disk galaxies (same structural parameters), this gives $r_0 \propto M_{\text{bar}}^{1/2}$, from which $v_f^4 = (GM_{\text{tot}}\,\alpha_t/r_0)^2 \propto M_{\text{bar}}$.

Caveat: The scaling $r_0 \propto M_{\text{bar}}^{1/2}$ is a structural argument that applies to galaxies with self-similar morphology. A first-principles derivation of the coherence scale r_0 from the RS Hamiltonian, and a quantitative prediction of the BTFR normalization A (not just the slope 4), are identified open problems. The qualitative BTFR slope is reproduced; the normalization requires additional calibration. □

Remark 7.3 (Radial Acceleration Relation — Structural Account)*.* The ILG model provides a structural account of the empirical Radial Acceleration Relation (RAR) [10]:

$$g_{\text{obs}} = \frac{g_{\text{bar}}}{1 - e^{-\sqrt{g_{\text{bar}}/g_\dagger}}}, \tag{10}$$

where $g_\dagger \approx 1.2 \times 10^{-10}$ m/s^2. In ILG, the acceleration scale $g_\dagger$ is not a free parameter but is set by α_t and the typical coherence length r_0 for disk galaxies: $g_\dagger \sim GM_{\text{disk}}\alpha_t/r_0^2$. This connects the "MOND acceleration scale" to the φ-ladder structure. A rigorous derivation of the precise RAR functional form from ILG dynamics is an identified open problem; the structural account shows that the scale $g_\dagger$ is not free, but does not yet derive the exponential factor.

8 Gravitational Lensing

Proposition 8.1 (Lensing from Substrate Coherence)**.** *The effective mass distribution for lensing includes the coherence contribution:*

$$\Sigma_{\text{eff}}(\mathbf{x}) = \Sigma_{\text{bar}}(\mathbf{x}) + \Sigma_C(\mathbf{x}), \tag{11}$$

where Σ_C is the surface density equivalent of the substrate coherence field.

Remark 8.2. Because coherence is enhanced in dense environments (galaxies, clusters) and reduced in voids, the ILG model produces lensing maps that mimic dark matter halos—without any particle.

9 The Bullet Cluster

Proposition 9.1 (Bullet Cluster Offset)**.** *The Bullet Cluster (1E 0657−56) [7] shows a striking offset: X-ray gas (baryonic) peaks between the two subclusters, while the gravitational lensing map peaks at the galaxy concentrations.*

In CDM, this is explained by collisionless dark matter halos that pass through each other while the gas is slowed by ram pressure. MOND struggles because it ties all dynamics to baryons [12].

Proposition 9.2 (ILG Explanation — Structural Account)**.** *In ILG, the substrate coherence follows the dominant mass component—the stellar content of the galaxies. During the collision:*

1. *Gas is stripped by ram pressure and displaced.*
2. *Coherence remains concentrated where the galaxies are, because coherence is sourced by the total J-cost density, which is dominated by the stellar (not gas) component.*
3. *The lensing map traces the coherence (galaxy) distribution, not the gas.*

No dark matter particle is required.

10 Ultra-Diffuse Galaxies

Proposition 10.1 (UDG Diversity)**.** *Ultra-diffuse galaxies (UDGs) span a range from dark-matter-rich (e.g., Dragonfly 44 [13]) to dark-matter-poor (e.g., NGC 1052-DF2 [14]).*

Remark 10.2. In CDM, it is difficult to explain dark-matter-free galaxies without invoking tidal stripping or formation scenarios [15]. In ILG, the substrate coherence varies naturally with environment:

- **Cluster UDGs**: Embedded in high-coherence environments, they acquire large effective dark matter fractions (Dragonfly 44-like).
- **Field UDGs**: In low-coherence environments, the tidal enhancement is weak, giving minimal dark matter effects (DF2-like).

The diversity of UDG dark matter fractions is a natural prediction, not a fine-tuning problem.

11 Comparison: CDM, MOND, and ILG

Observable	**ΛCDM**	**MOND**	**RS/ILG**
DM particle	WIMP/axion	None	None
DM substance	Yes	No	Yes (ledger shadows)
Rotation curves	Tuned halos	a_0	α_t
BTFR	Emergent	Built-in	Derived
RAR	Requires tuning	Built-in	Derived
UDG diversity	Difficult	Difficult	Natural
Bullet Cluster	Explained	Difficult	Explained
Ω_{dm}	Fitted	N/A	Derived
Direct detection	Predicted	N/A	Null predicted
Free parameters	ρ_s, r_s per halo	a_0	None

Remark 11.1. The key distinction: RS/ILG is *not* modified gravity. Gravity follows standard GR (with emergent corrections from the substrate). The dark sector is a real physical component (odd-phase ledger entries), but it is not a new particle species. This positions RS between CDM (which introduces new particles) and MOND (which modifies gravity).

12 Predictions and Falsifiers

1. **No particle detection**: Null results in all WIMP, axion, and sterile neutrino searches (XENONnT, LZ, PandaX, ADMX, any future experiment). Any confirmed detection of a dark matter particle would falsify the ledger shadow model.

2. **Universal α_t**: Galaxy rotation curves should be fit by the ILG model with a single universal $\alpha_t = 0.191$, not per-galaxy halo parameters. Inconsistency with this value would falsify ILG.

3. $\Omega_{\mathrm{dm}} \approx 0.265$: The dark matter density parameter is derived, not fitted. Significant deviation from this value (beyond systematic uncertainties) would falsify the 8-tick partition mechanism.

4. **UDG diversity**: The model predicts a continuous distribution of dark matter fractions in UDGs, correlated with environment.

5. **No self-annihilation signal**: Since dark matter is not a particle–antiparticle pair, there is no annihilation cross section. Indirect detection searches (Fermi-LAT, HESS, CTA) should find null results.

6. **CMB consistency**: The dark sector imprints on the CMB through its gravitational influence, with the same Ω_{dm} that determines structure formation.

13 Conclusion

Dark matter in RS is neither a particle nor modified gravity. It is a structural feature of the recognition ledger: the odd-phase sector of the 8-tick epoch. These "ledger shadows" gravitate through J-cost coupling but do not interact electromagnetically. The ILG model describes the resulting gravitational dynamics with a single derived parameter $\alpha_t = \frac{1}{2}(1 - \varphi^{-1}) \approx 0.191$, producing flat rotation curves, the BTFR, the RAR, and the Bullet Cluster offset—all without free parameters. The dark matter fraction $\Omega_{\rm dm} \approx 0.265$ is derived from the edge-channel geometry of the $D = 3$ cube.

The model makes sharp, falsifiable predictions: no dark matter particle will be detected, and all galaxy rotation curves will be fit by the universal α_t.

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